

Question ID affb2315

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: affb2315

SAT4 M 3.1

There are **55** students in Spanish club. A sample of the Spanish club students was selected at random and asked whether they intend to enroll in a new study program. Of those surveyed, **20%** responded that they intend to enroll in the study program. Based on this survey, which of the following is the best estimate of the total number of Spanish club students who intend to enroll in the study program?

- A. **11**
- B. **20**
- C. **44**
- D. **55**

ID: affb2315 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that **20%** of the students surveyed responded that they intend to enroll in the study program. Therefore, the proportion of students in Spanish club who intend to enroll in the study program, based on the survey, is **0.20**. Since there are **55** total students in Spanish club, the best estimate for the total number of these students who intend to enroll in the study program is **55(0.20)**, or **11**.

Choice B is incorrect. This is the best estimate for the percentage, rather than the total number, of students in Spanish club who intend to enroll in the study program.

Choice C is incorrect. This is the best estimate for the total number of Spanish club students who do not intend to enroll in the study program.

Choice D is incorrect. This is the total number of students in Spanish club.

Question Difficulty: Easy

Question ID ebf8d2b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: ebf8d2b7

SAT4 M 3.2

A machine makes large boxes or small boxes, one at a time, for a total of **700** minutes each day. It takes the machine **10** minutes to make a large box or **5** minutes to make a small box. Which equation represents the possible number of large boxes, x , and small boxes, y , the machine can make each day?

- A. $5x + 10y = 700$
- B. $10x + 5y = 700$
- C. $(x + y)(10 + 5) = 700$
- D. $(10 + x)(5 + y) = 700$

ID: ebf8d2b7 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that it takes the machine **10** minutes to make a large box. It's also given that x represents the possible number of large boxes the machine can make each day. Multiplying **10** by x gives $10x$, which represents the amount of time spent making large boxes. It's given that it takes the machine **5** minutes to make a small box. It's also given that y represents the possible number of small boxes the machine can make each day. Multiplying **5** by y gives $5y$, which represents the amount of time spent making small boxes. Combining the amount of time spent making x large boxes and y small boxes yields $10x + 5y$. It's given that the machine makes boxes for a total of **700** minutes each day. Therefore $10x + 5y = 700$ represents the possible number of large boxes, x , and small boxes, y , the machine can make each day.

Choice A is incorrect and may result from associating the time of **10** minutes with small, rather than large, boxes and the time of **5** minutes with large, rather than small, boxes.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

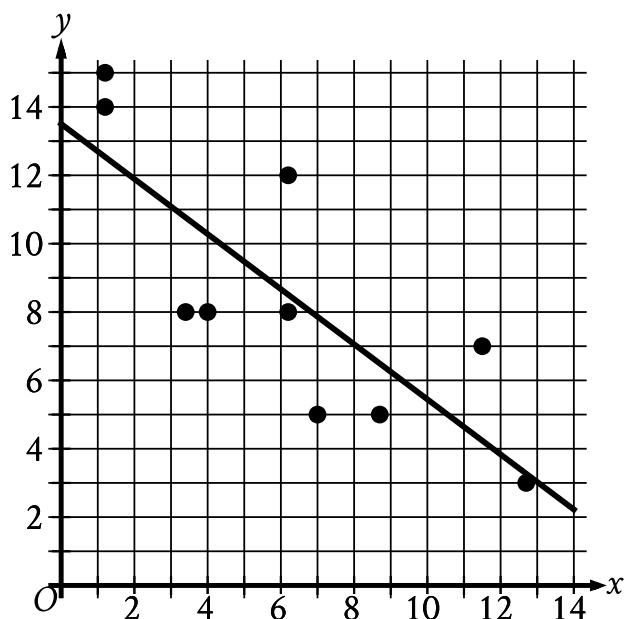
Question ID 8baf2118

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 8baf2118

SAT4 M 3.3

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

- A. $y = 13.5 + 0.8x$
- B. $y = 13.5 - 0.8x$
- C. $y = -13.5 + 0.8x$
- D. $y = -13.5 - 0.8x$

ID: 8baf2118 Answer

Correct Answer: B

Rationale

Choice B is correct. The line of best fit shown intersects the y -axis at a positive y -value and has a negative slope. The graph of an equation of the form $y = a + bx$, where a and b are constants, intersects the y -axis at a y -value of a and has a slope of b . Of the given choices, only choice B represents a line that intersects the y -axis at a positive y -value, 13.5 , and has a negative slope, -0.8 .

Choice A is incorrect. This equation represents a line that has a positive slope, not a negative slope.

Choice C is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value, and has a positive slope, not a negative slope.

Choice D is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value.

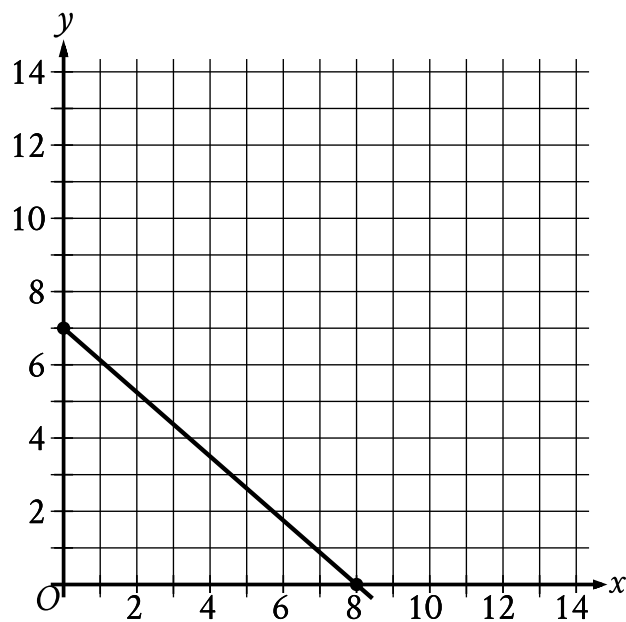
Question Difficulty: Easy

Question ID 9d0396d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■ ■ ■

ID: 9d0396d4

SAT4 M 3.4



The point with coordinates $(d, 4)$ lies on the line shown. What is the value of d ?

- A. $\frac{7}{2}$
- B. $\frac{26}{7}$
- C. $\frac{24}{7}$
- D. $\frac{27}{8}$

ID: 9d0396d4 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given from the graph that the points $(0, 7)$ and $(8, 0)$ lie on the line. For two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(0, 7)$ for (x_1, y_1) and $(8, 0)$ for (x_2, y_2) in this formula, the slope of the line can be calculated as $m = \frac{0 - 7}{8 - 0}$, or $m = -\frac{7}{8}$. It's also given that the point $(d, 4)$ lies on the line. Substituting $(d, 4)$ for (x_1, y_1) , $(8, 0)$ for (x_2, y_2) , and $-\frac{7}{8}$ for m in the slope formula yields $-\frac{7}{8} = \frac{0 - 4}{8 - d}$, or $-\frac{7}{8} = \frac{-4}{8 - d}$. Multiplying both sides of this equation by $8 - d$ yields $-\frac{7}{8}(8 - d) = -4$. Expanding the left-hand side of this equation yields $-7 + \frac{7}{8}d = -4$. Adding 7 to both sides of this equation yields $\frac{7}{8}d = 3$. Multiplying both sides of this equation by $\frac{8}{7}$ yields $d = \frac{24}{7}$. Thus, the value of d is $\frac{24}{7}$.

Question ID 44076c7d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 44076c7d

SAT4 M 3.5

x	$f(x)$
-1	10
0	14
1	20

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

- A. $f(x) = 3x^2 + 3x + 14$
- B. $f(x) = 5x^2 + x + 14$
- C. $f(x) = 9x^2 - x + 14$
- D. $f(x) = x^2 + 5x + 14$

ID: 44076c7d Answer

Correct Answer: D

Rationale

Choice D is correct. The equation of a quadratic function can be written in the form $f(x) = a(x - h)^2 + k$, where a , h , and k are constants. It's given in the table that when $x = -1$, the corresponding value of $f(x)$ is 10 . Substituting -1 for x and 10 for $f(x)$ in the equation $f(x) = a(x - h)^2 + k$ gives $10 = a(-1 - h)^2 + k$, which is equivalent to $10 = a(1 + 2h + h^2) + k$, or $10 = a + 2ah + ah^2 + k$. It's given in the table that when $x = 0$, the corresponding value of $f(x)$ is 14 . Substituting 0 for x and 14 for $f(x)$ in the equation $f(x) = a(x - h)^2 + k$ gives $14 = a(0 - h)^2 + k$, or $14 = ah^2 + k$. It's given in the table that when $x = 1$, the corresponding value of $f(x)$ is 20 . Substituting 1 for x and 20 for $f(x)$ in the equation $f(x) = a(x - h)^2 + k$ gives $20 = a(1 - h)^2 + k$, which is equivalent to $20 = a(1 - 2h + h^2) + k$, or $20 = a - 2ah + ah^2 + k$. Adding $20 = a - 2ah + ah^2 + k$ to the equation $10 = a + 2ah + ah^2 + k$ gives $30 = 2a + 2ah^2 + 2k$. Dividing both sides of this equation by 2 gives $15 = a + ah^2 + k$. Since $14 = ah^2 + k$, substituting 14 for $ah^2 + k$ into the equation $15 = a + ah^2 + k$ gives $15 = a + 14$. Subtracting 14 from both sides of this equation gives $a = 1$. Substituting 1 for a in the equations $14 = ah^2 + k$ and $20 = ah^2 - 2ah + a + k$ gives $14 = h^2 + k$ and $20 = 1 - 2h + h^2 + k$, respectively. Since $14 = h^2 + k$, substituting 14 for $h^2 + k$ in the equation $20 = 1 - 2h + h^2 + k$ gives $20 = 1 - 2h + 14$, or $20 = 15 - 2h$. Subtracting 15 from both sides of this equation gives $5 = -2h$. Dividing both sides of this equation by -2 gives $-\frac{5}{2} = h$. Substituting $-\frac{5}{2}$ for h into the equation $14 = h^2 + k$ gives $14 = \left(-\frac{5}{2}\right)^2 + k$, or $14 = \frac{25}{4} + k$. Subtracting $\frac{25}{4}$ from both sides of this equation gives $\frac{31}{4} = k$. Substituting 1 for a , $-\frac{5}{2}$ for h , and $\frac{31}{4}$ for k in the equation $f(x) = a(x - h)^2 + k$ gives $f(x) = \left(x + \frac{5}{2}\right)^2 + \frac{31}{4}$, which is equivalent to $f(x) = x^2 + 5x + \frac{25}{4} + \frac{31}{4}$, or $f(x) = x^2 + 5x + 14$. Therefore, $f(x) = x^2 + 5x + 14$ defines f .

Choice A is incorrect. If $f(x) = 3x^2 + 3x + 14$, then when $x = -1$, the corresponding value of $f(x)$ is 14, not 10.

Choice B is incorrect. If $f(x) = 5x^2 + x + 14$, then when $x = -1$, the corresponding value of $f(x)$ is 18, not 10.

Choice C is incorrect. If $f(x) = 9x^2 - x + 14$, then when $x = -1$, the corresponding value of $f(x)$ is 24, not 10, and when $x = 1$, the corresponding value of $f(x)$ is 22, not 20.

Question Difficulty: Medium

Question ID 0cadb20e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0cadb20e

SAT4 M 3.6

The function f is defined by $f(x) = \frac{x+15}{5}$, and $f(a) = 10$, where a is a constant. What is the value of a ?

- A. 5
- B. 10
- C. 35
- D. 65

ID: 0cadb20e Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $f(x) = \frac{x+15}{5}$ and $f(a) = 10$, where a is a constant. Therefore, for the given function f , when $x = a$, $f(x) = 10$. Substituting a for x and 10 for $f(x)$ in the given function f yields $10 = \frac{a+15}{5}$. Multiplying both sides of this equation by 5 yields $50 = a + 15$. Subtracting 15 from both sides of this equation yields $35 = a$. Therefore, the value of a is 35.

Choice A is incorrect. This is the value of a if $f(a) = 4$.

Choice B is incorrect. This is the value of a if $f(a) = 5$.

Choice D is incorrect. This is the value of a if $f(a) = 16$.

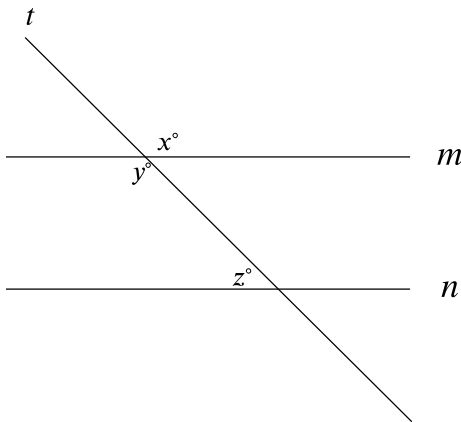
Question Difficulty: Medium

Question ID 2adbf1b1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 2adbf1b1

SAT4 M 3.7



Note: Figure not drawn to scale.

In the figure, lines *m* and *n* are parallel. If $x = 6k + 13$ and $y = 8k - 29$, what is the value of z ?

- A. 3
- B. 21
- C. 41
- D. 139

ID: 2adbf1b1 Answer

Correct Answer: C

Rationale

Choice C is correct. Vertical angles, which are angles that are opposite each other when two lines intersect, are congruent. The figure shows that lines *t* and *m* intersect. It follows that the angle with measure x° and the angle with measure y° are vertical angles, so $x = y$. It's given that $x = 6k + 13$ and $y = 8k - 29$. Substituting $6k + 13$ for x and $8k - 29$ for y in the equation $x = y$ yields $6k + 13 = 8k - 29$. Subtracting $6k$ from both sides of this equation yields $13 = 2k - 29$. Adding 29 to both sides of this equation yields $42 = 2k$, or $2k = 42$. Dividing both sides of this equation by 2 yields $k = 21$. It's given that lines *m* and *n* are parallel, and the figure shows that lines *m* and *n* are intersected by a transversal, line *t*. If two parallel lines are intersected by a transversal, then the same-side interior angles are supplementary. It follows that the same-side interior angles with measures y° and z° are supplementary, so $y + z = 180$. Substituting $8k - 29$ for y in this equation yields $8k - 29 + z = 180$. Substituting 21 for k in this equation yields $8(21) - 29 + z = 180$, or $139 + z = 180$. Subtracting 139 from both sides of this equation yields $z = 41$. Therefore, the value of z is 41 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of k , not z .

Choice D is incorrect. This is the value of x or y , not z .

Question Difficulty: Medium

Question ID 01682aa5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: 01682aa5

SAT4 M 3.8

Line p is defined by $2y + 18x = 9$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

- A. -9
- B. $-\frac{1}{9}$
- C. $\frac{1}{9}$
- D. 9

ID: 01682aa5 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that line r is perpendicular to line p in the xy -plane. This means that the slope of line r is the negative reciprocal of the slope of line p . If the equation for line p is rewritten in slope-intercept form $y = mx + b$, where m and b are constants, then m is the slope of the line and $(0, b)$ is its y -intercept. Subtracting $18x$ from both sides of the equation $2y + 18x = 9$ yields $2y = -18x + 9$. Dividing both sides of this equation by 2 yields $y = -9x + \frac{9}{2}$. It follows that the slope of line p is -9 . The negative reciprocal of a number is -1 divided by the number. Therefore, the negative reciprocal of -9 is $\frac{-1}{-9}$, or $\frac{1}{9}$. Thus, the slope of line r is $\frac{1}{9}$.

Choice A is incorrect. This is the slope of line p , not line r .

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of the slope of line p .

Choice D is incorrect. This is the negative, not the negative reciprocal, of the slope of line p .

Question Difficulty: Medium

Question ID 69f6717f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	■ ■ ■

ID: 69f6717f

SAT4 M 3.9

A sample of oak has a density of **807** kilograms per cubic meter. The sample is in the shape of a cube, where each edge has a length of **0.90** meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

- A. **588**
- B. **726**
- C. **897**
- D. **1,107**

ID: 69f6717f Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the sample is in the shape of a cube with edge lengths of **0.9** meters. Therefore, the volume of the sample is **0.90³**, or **0.729**, cubic meters. It's also given that the sample has a density of **807** kilograms per **1** cubic meter. Therefore, the mass of this sample is **0.729 cubic meters** $\left(\frac{807 \text{ kilograms}}{1 \text{ cubic meter}}\right)$, or **588.303** kilograms. Rounding this mass to the nearest whole number gives **588** kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is **588**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 2d1614a1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■ ■ ■

ID: 2d1614a1

SAT4 M 3.10

$$P(t) = 290(1.04)^{\left(\frac{4}{6}\right)t}$$

The function P models the population, in thousands, of a certain city t years after **2005**. According to the model, the population is predicted to increase by $n\%$ every **18** months. What is the value of n ?

- A. **0.38**
- B. **1.04**
- C. **4**
- D. **6**

ID: 2d1614a1 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the function P models the population of the city t years after **2005**. Since there are **12** months in a year, **18** months is equivalent to $\frac{18}{12}$ years. Therefore, the expression $\frac{18}{12}x$ can represent the number of years in x **18**-month periods. Substituting $\frac{18}{12}x$ for t in the given equation yields $P\left(\frac{18}{12}x\right) = 290(1.04)^{\left(\frac{4}{6}\right)\left(\frac{18}{12}x\right)}$, which is equivalent to $P\left(\frac{18}{12}x\right) = 290(1.04)^x$. Therefore, for each **18**-month period, the predicted population of the city is **1.04** times, or **104%** of, the previous population. This means that the population is predicted to increase by **4%** every **18** months.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. Each year, the predicted population of the city is **1.04** times the previous year's predicted population, which is not the same as an increase of **1.04%**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 7d5d1b32

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	■ ■ ■

ID: 7d5d1b32

SAT4 M 3.11

$$2(kx - n) = -\frac{28}{15}x - \frac{36}{19}$$

In the given equation, k and n are constants and $n > 1$. The equation has no solution. What is the value of k ?

ID: 7d5d1b32 Answer

Correct Answer: -.9333, -14/15

Rationale

The correct answer is $-\frac{14}{15}$. A linear equation in the form $ax + b = cx + d$ has no solution only when the coefficients of x on each side of the equation are equal and the constant terms are not equal. Dividing both sides of the given equation by 2 yields $kx - n = -\frac{28}{30}x - \frac{36}{38}$, or $kx - n = -\frac{14}{15}x - \frac{18}{19}$. Since it's given that the equation has no solution, the coefficient of x on both sides of this equation must be equal, and the constant terms on both sides of this equation must not be equal. Since $\frac{18}{19} < 1$, and it's given that $n > 1$, the second condition is true. Thus, k must be equal to $-\frac{14}{15}$. Note that -14/15, -.9333, and -0.933 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 3d73a58b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	■ ■ ■

ID: 3d73a58b

SAT4 M 3.12

A gift shop buys souvenirs at a wholesale price of **7.00** dollars each and resells them each at a retail price that is **290%** of the wholesale price. At the end of the season, any remaining souvenirs are marked at a discounted price that is **80%** off the retail price. What is the discounted price of each remaining souvenir, in dollars?

ID: 3d73a58b Answer

Correct Answer: 203/50, 4.06

Rationale

The correct answer is **4.06**. It's given that the retail price is **290%** of the wholesale price of **\$7.00**. Thus, the retail price is **\$7.00($\frac{290}{100}$)**, which is equivalent to **\$7.00(2.9)**, or **\$20.30**. It's also given that the discounted price is **80%** off the retail price. Thus, the discounted price is **\$20.30($1 - \frac{80}{100}$)**, which is equivalent to **\$20.30(0.20)**, or **\$4.06**.

Question Difficulty: Hard

Question ID ebb717ab

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: ebb717ab

SAT4 M 3.13

$$x^2 - 34x + c = 0$$

In the given equation, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n ?

ID: ebb717ab Answer

Correct Answer: 289

Rationale

The correct answer is **289**. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solutions when the value of the discriminant, $b^2 - 4ac$, is less than 0. In the given equation, $x^2 - 34x + c = 0$, $a = 1$ and $b = -34$. Therefore, the discriminant of the given equation can be expressed as $(-34)^2 - 4(1)(c)$, or $1,156 - 4c$. It follows that the given equation has no real solutions when $1,156 - 4c < 0$. Adding $4c$ to both sides of this inequality yields $1,156 < 4c$. Dividing both sides of this inequality by 4 yields $289 < c$, or $c > 289$. It's given that the equation $x^2 - 34x + c = 0$ has no real solutions when $c > n$. Therefore, the least possible value of n is **289**.

Question Difficulty: Hard

Question ID 2a59eb45

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	■ ■ ■

ID: 2a59eb45

SAT4 M 3.14

Data set A consists of the heights of **75** buildings and has a mean of **32** meters. Data set B consists of the heights of **50** buildings and has a mean of **62** meters. Data set C consists of the heights of the **125** buildings from data sets A and B. What is the mean, in meters, of data set C?

ID: 2a59eb45 Answer

Correct Answer: 44

Rationale

The correct answer is **44**. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of **75** buildings and has a mean of **32** meters. This can be represented by the equation $\frac{x}{75} = 32$, where x represents the sum of the heights of the buildings, in meters, in data set A. Multiplying both sides of this equation by **75** yields $x = 75(32)$, or $x = 2,400$ meters. Therefore, the sum of the heights of the buildings in data set A is **2,400** meters. It's also given that data set B consists of the heights of **50** buildings and has a mean of **62** meters. This can be represented by the equation $\frac{y}{50} = 62$, where y represents the sum of the heights of the buildings, in meters, in data set B. Multiplying both sides of this equation by **50** yields $y = 50(62)$, or $y = 3,100$ meters. Therefore, the sum of the heights of the buildings in data set B is **3,100** meters. Since it's given that data set C consists of the heights of the **125** buildings from data sets A and B, it follows that the mean of data set C is the sum of the heights of the buildings, in meters, in data sets A and B divided by the number of buildings represented in data sets A and B, or $\frac{2,400+3,100}{125}$, which is equivalent to **44** meters. Therefore, the mean, in meters, of data set C is **44**.

Question Difficulty: Hard

Question ID ffdbcad4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	■ ■ ■

ID: ffdbcad4

SAT4 M 3.15

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as $(hx + k)(x + j)$, where h , k , and j are integer constants. Which of the following must be an integer?

- A. $\frac{b}{h}$
- B. $\frac{b}{k}$
- C. $\frac{45}{h}$
- D. $\frac{45}{k}$

ID: ffdbcad4 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $4x^2 + bx - 45$ can be rewritten as $(hx + k)(x + j)$. The expression $(hx + k)(x + j)$ can be rewritten as $hx^2 + jhx + kx + kj$, or $hx^2 + (jh + k)x + kj$. Therefore, $hx^2 + (jh + k)x + kj$ is equivalent to $4x^2 + bx - 45$. It follows that $kj = -45$. Dividing each side of this equation by k yields $j = \frac{-45}{k}$. Since j is an integer, $-\frac{45}{k}$ must be an integer. Therefore, $\frac{45}{k}$ must also be an integer.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 9cb9beec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: 9cb9beec

SAT4 M 3.16

$$\begin{aligned}y &= -1.50 \\ y &= x^2 + 8x + a\end{aligned}$$

In the given system of equations, a is a positive constant. The system has exactly one distinct real solution. What is the value of a ?

ID: 9cb9beec Answer

Correct Answer: 14.5, 29/2

Rationale

The correct answer is $\frac{29}{2}$. According to the first equation in the given system, the value of y is -1.5 . Substituting -1.5 for y in the second equation in the given system yields $-1.5 = x^2 + 8x + a$. Adding 1.5 to both sides of this equation yields $0 = x^2 + 8x + a + 1.5$. If the given system has exactly one distinct real solution, it follows that $0 = x^2 + 8x + a + 1.5$ has exactly one distinct real solution. A quadratic equation in the form $0 = px^2 + qx + r$, where p , q , and r are constants, has exactly one distinct real solution if and only if the discriminant, $q^2 - 4pr$, is equal to 0 . The equation $0 = x^2 + 8x + a + 1.5$ is in this form, where $p = 1$, $q = 8$, and $r = a + 1.5$. Therefore, the discriminant of the equation $0 = x^2 + 8x + a + 1.5$ is $(8)^2 - 4(1)(a + 1.5)$, or $58 - 4a$. Setting the discriminant equal to 0 to solve for a yields $58 - 4a = 0$. Adding $4a$ to both sides of this equation yields $58 = 4a$. Dividing both sides of this equation by 4 yields $\frac{58}{4} = a$, or $\frac{29}{2} = a$. Therefore, if the given system of equations has exactly one distinct real solution, the value of a is $\frac{29}{2}$. Note that 29/2 and 14.5 are examples of ways to enter a correct answer.

Question Difficulty: Hard

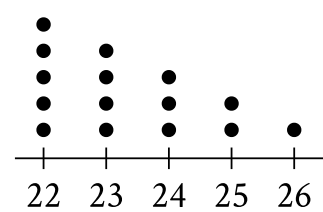
Question ID 4626102e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	■ ■ ■

ID: 4626102e

SAT4 M 3.17

Data Set A



- The dot plot represents the **15** values in data set A. Data set B is created by adding **56** to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?
- A. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.
 - B. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
 - C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
 - D. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

ID: 4626102e Answer

Correct Answer: C

Rationale

Choice C is correct. The median of a data set with an odd number of values, in ascending or descending order, is the middle value of the data set, and the range of a data set is the positive difference between the maximum and minimum values in the data set. Since the dot plot shown gives the values in data set A in ascending order and there are **15** values in the data set, the eighth value in data set A, **23**, is the median. The maximum value in data set A is **26** and the minimum value is **22**, so the range of data set A is **26 – 22**, or **4**. It's given that data set B is created by adding **56** to each of the values in data set A. Increasing each of the **15** values in data set A by **56** will also increase its median value by **56** making the median of data set B **79**. Increasing each value of data set A by **56** does not change the range, since the maximum value of data set B is **26 + 56**, or **82**, and the minimum value is **22 + 56**, or **78**, making the range of data set B **82 – 78**, or **4**. Therefore, the median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

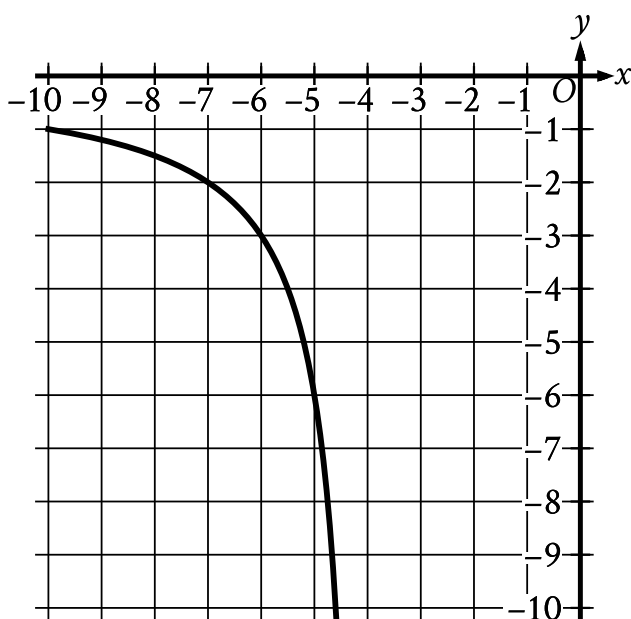
Choice B is incorrect and may result from conceptual or calculation errors.

Question ID 4d037075

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■ ■ ■

ID: 4d037075

SAT4 M 3.18



The rational function f is defined by an equation in the form $f(x) = \frac{a}{x+b}$, where a and b are constants. The partial graph of $y = f(x)$ is shown. If $g(x) = f(x+4)$, which equation could define function g ?

- A. $g(x) = \frac{6}{x}$
- B. $g(x) = \frac{6}{x+4}$
- C. $g(x) = \frac{6}{x+8}$
- D. $g(x) = \frac{6(x+4)}{x+4}$

ID: 4d037075 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $f(x) = \frac{a}{x+b}$ and that the graph shown is a partial graph of $y = f(x)$. Substituting y for $f(x)$ in the equation $f(x) = \frac{a}{x+b}$ yields $y = \frac{a}{x+b}$. The graph passes through the point $(-7, -2)$. Substituting -7 for x and -2 for y in the equation $y = \frac{a}{x+b}$ yields $-2 = \frac{a}{-7+b}$. Multiplying each side of this equation by $-7+b$ yields $-2(-7+b) = a$, or $14 - 2b = a$. The graph also passes through the point $(-5, -6)$. Substituting -5 for x and -6 for y in the equation $y = \frac{a}{x+b}$ yields $-6 = \frac{a}{-5+b}$. Multiplying each side of this equation by $-5+b$ yields $-6(-5+b) = a$, or $30 - 6b = a$. Substituting $14 - 2b$ for a in this equation yields $30 - 6b = 14 - 2b$. Adding $6b$ to each side of this equation yields $30 = 14 + 4b$.

Subtracting 14 from each side of this equation yields $16 = 4b$. Dividing each side of this equation by 4 yields $4 = b$. Substituting 4 for b in the equation $14 - 2b = a$ yields $14 - 2(4) = a$, or $6 = a$. Substituting 6 for a and 4 for b in the equation $f(x) = \frac{a}{x+b}$ yields $f(x) = \frac{6}{x+4}$. It's given that $g(x) = f(x+4)$. Substituting $x+4$ for x in the equation $f(x) = \frac{6}{x+4}$ yields $f(x+4) = \frac{6}{x+4+4}$, which is equivalent to $f(x+4) = \frac{6}{x+8}$. It follows that $g(x) = \frac{6}{x+8}$.

Choice A is incorrect. This could define function g if $g(x) = f(x-4)$.

Choice B is incorrect. This could define function g if $g(x) = f(x)$.

Choice D is incorrect. This could define function g if $g(x) = f(x) \cdot (x+4)$.

Question Difficulty: Hard

Question ID 2c05d312

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: 2c05d312

SAT4 M 3.19

$$57x^2 + (57b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab , where k is a constant. What is the value of k ?

- A. $\frac{1}{57}$
- B. $\frac{1}{19}$
- C. 1
- D. 57

ID: 2c05d312 Answer

Correct Answer: A

Rationale

Choice A is correct. The left-hand side of the given equation is the expression $57x^2 + (57b + a)x + ab$. Applying the distributive property to this expression yields $57x^2 + 57bx + ax + ab$. Since the first two terms of this expression have a common factor of $57x$ and the last two terms of this expression have a common factor of a , this expression can be rewritten as $57x(x + b) + a(x + b)$. Since the two terms of this expression have a common factor of $(x + b)$, it can be rewritten as $(x + b)(57x + a)$. Therefore, the given equation can be rewritten as $(x + b)(57x + a) = 0$. By the zero product property, it follows that $x + b = 0$ or $57x + a = 0$. Subtracting b from both sides of the equation $x + b = 0$ yields $x = -b$. Subtracting a from both sides of the equation $57x + a = 0$ yields $57x = -a$. Dividing both sides of this equation by 57 yields $x = \frac{-a}{57}$. Therefore, the solutions to the given equation are $-b$ and $\frac{-a}{57}$. It follows that the product of the solutions of the given equation is $(-b)(\frac{-a}{57})$, or $\frac{ab}{57}$. It's given that the product of the solutions of the given equation is kab . It follows that $\frac{ab}{57} = kab$, which can also be written as $ab(\frac{1}{57}) = ab(k)$. It's given that a and b are positive constants. Therefore, dividing both sides of the equation $ab(\frac{1}{57}) = ab(k)$ by ab yields $\frac{1}{57} = k$. Thus, the value of k is $\frac{1}{57}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 76c73dbf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	■ ■ ■

ID: 76c73dbf

SAT4 M 3.20

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy -plane is a circle. What is the length of the circle's radius?

ID: 76c73dbf Answer

Correct Answer: 10

Rationale

The correct answer is **10**. It's given that the graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy -plane is a circle. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the length of the radius of the circle is r . The term $(x - h)^2$ in this equation can be obtained by adding the square of half the coefficient of x to both sides of the given equation to complete the square. The coefficient of x is **1**. Half the coefficient of x is $\frac{1}{2}$. The square of half the coefficient of x is $\frac{1}{4}$. Adding $\frac{1}{4}$ to each side of $(x^2 + x) + (y^2 + y) = \frac{199}{2}$ yields $(x^2 + x + \frac{1}{4}) + (y^2 + y) = \frac{199}{2} + \frac{1}{4}$, or $(x + \frac{1}{2})^2 + (y^2 + y) = \frac{199}{2} + \frac{1}{4}$. Similarly, the term $(y - k)^2$ can be obtained by adding the square of half the coefficient of y to both sides of this equation, which yields $(x + \frac{1}{2})^2 + (y^2 + y + \frac{1}{4}) = \frac{199}{2} + \frac{1}{4} + \frac{1}{4}$, or $(x + \frac{1}{2})^2 + (y + \frac{1}{2})^2 = \frac{199}{2} + \frac{1}{4} + \frac{1}{4}$. This equation is equivalent to $(x + \frac{1}{2})^2 + (y + \frac{1}{2})^2 = 100$, or $(x + \frac{1}{2})^2 + (y + \frac{1}{2})^2 = 10^2$. Therefore, the length of the circle's radius is **10**.

Question Difficulty: Hard

Question ID f243c383

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	■ ■ ■

ID: f243c383

SAT4 M 3.21

Two identical rectangular prisms each have a height of **90 centimeters (cm)**. The base of each prism is a square, and the surface area of each prism is $K \text{ cm}^2$. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K \text{ cm}^2$. What is the side length, in **cm**, of each square base?

- A. 4
- B. 8
- C. 9
- D. 16

ID: f243c383 Answer

Correct Answer: B

Rationale

Choice B is correct. Let x represent the side length, in **cm**, of each square base. If the two prisms are glued together along a square base, the resulting prism has a surface area equal to twice the surface area of one of the prisms, minus the area of the two square bases that are being glued together, which yields $2K - 2x^2 \text{ cm}^2$. It's given that this resulting surface area is equal to $\frac{92}{47}K \text{ cm}^2$, so $2K - 2x^2 = \frac{92}{47}K$. Subtracting $\frac{92}{47}K$ from both sides of this equation yields $2K - \frac{92}{47}K - 2x^2 = 0$. This equation can be rewritten by multiplying $2K$ on the left-hand side by $\frac{47}{47}$, which yields $\frac{94}{47}K - \frac{92}{47}K - 2x^2 = 0$, or $\frac{2}{47}K - 2x^2 = 0$. Adding $2x^2$ to both sides of this equation yields $\frac{2}{47}K = 2x^2$. Multiplying both sides of this equation by $\frac{47}{2}$ yields $K = 47x^2$. The surface area K , in cm^2 , of each rectangular prism is equivalent to the sum of the areas of the two square bases and the areas of the four lateral faces. Since the height of each rectangular prism is **90 cm** and the side length of each square base is $x \text{ cm}$, it follows that the area of each square base is $x^2 \text{ cm}^2$ and the area of each lateral face is $90x \text{ cm}^2$. Therefore, the surface area of each rectangular prism can be represented by the expression $2x^2 + 4(90x)$, or $2x^2 + 360x$. Substituting this expression for K in the equation $K = 47x^2$ yields $2x^2 + 360x = 47x^2$. Subtracting $2x^2$ and $360x$ from both sides of this equation yields $0 = 45x^2 - 360x$. Factoring x from the right-hand side of this equation yields $0 = x(45x - 360)$. Applying the zero product property, it follows that $x = 0$ and $45x - 360 = 0$. Adding 360 to both sides of the equation $45x - 360 = 0$ yields $45x = 360$. Dividing both sides of this equation by 45 yields $x = 8$. Since a side length of a rectangular prism can't be 0 , the length of each square base is **8 cm**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 92f812bb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 92f812bb

SAT4 M 3.22

In the xy -plane, a parabola has vertex $(9, -14)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants, which of the following could be the value of $a + b + c$?

- A. -23
- B. -19
- C. -14
- D. -12

ID: 92f812bb Answer

Correct Answer: D

Rationale

Choice D is correct. The equation of a parabola in the xy -plane can be written in the form $y = a(x - h)^2 + k$, where a is a constant and (h, k) is the vertex of the parabola. If a is positive, the parabola will open upward, and if a is negative, the parabola will open downward. It's given that the parabola has vertex $(9, -14)$. Substituting 9 for h and -14 for k in the equation $y = a(x - h)^2 + k$ gives $y = a(x - 9)^2 - 14$, which can be rewritten as $y = a(x - 9)(x - 9) - 14$, or $y = a(x^2 - 18x + 81) - 14$. Distributing the factor of a on the right-hand side of this equation yields $y = ax^2 - 18ax + 81a - 14$. Therefore, the equation of the parabola, $y = ax^2 - 18ax + 81a - 14$, can be written in the form $y = ax^2 + bx + c$, where $a = a$, $b = -18a$, and $c = 81a - 14$. Substituting $-18a$ for b and $81a - 14$ for c in the expression $a + b + c$ yields $(a) + (-18a) + (81a - 14)$, or $64a - 14$. Since the vertex of the parabola, $(9, -14)$, is below the x -axis, and it's given that the parabola intersects the x -axis at two points, the parabola must open upward. Therefore, the constant a must have a positive value. Setting the expression $64a - 14$ equal to the value in choice D yields $64a - 14 = -12$. Adding 14 to both sides of this equation yields $64a = 2$. Dividing both sides of this equation by 64 yields $a = \frac{2}{64}$, which is a positive value. Therefore, if the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants, the value of $a + b + c$ could be -12 .

Choice A is incorrect. If the equation of a parabola with a vertex at $(9, -14)$ is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants and $a + b + c = -23$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the x -axis at zero points, not two points.

Choice B is incorrect. If the equation of a parabola with a vertex at $(9, -14)$ is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants and $a + b + c = -19$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the x -axis at zero points, not two points.

Choice C is incorrect. If the equation of a parabola with a vertex at $(9, -14)$ is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants and $a + b + c = -14$, then the value of a will be 0, which is inconsistent with the equation of a parabola.

Question Difficulty: Hard